

Math 526, S16
Exam 1

Name: *Answer*
SSN: xxx-xx-

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	20	
2	20	
3	24	
4	20	
5	20	
Total	104	

Good Luck !

Problem 1. (20 points) Given two vectors in the three dimensional space $\mathbf{v} = (3, \sqrt{3}, 2\sqrt{6})$ and $\mathbf{w} = (0, -\frac{2\sqrt{3}}{3}, \frac{2\sqrt{6}}{3})$.

(a) Find the length of $\mathbf{v}$ and $\mathbf{w}$.

$$|\mathbf{v}| = \sqrt{3^2 + (\sqrt{3})^2 + (2\sqrt{6})^2} = \sqrt{36} = 6$$

$$|\mathbf{w}| = \sqrt{0^2 + (-\frac{2\sqrt{3}}{3})^2 + (\frac{2\sqrt{6}}{3})^2} = \sqrt{0 + \frac{4}{3} + \frac{8}{3}} = \sqrt{4} = 2$$

(b) Find the angle θ between $\mathbf{v}$ and $\mathbf{w}$.

$$\begin{aligned} \mathbf{v} \cdot \mathbf{w} &= (3, \sqrt{3}, 2\sqrt{6}) \cdot (0, -\frac{2\sqrt{3}}{3}, \frac{2\sqrt{6}}{3}) \\ &= 3 \cdot 0 + \sqrt{3} \cdot (-\frac{2\sqrt{3}}{3}) + 2\sqrt{6} \cdot \frac{2\sqrt{6}}{3} = -\frac{2 \cdot 3}{3} + \frac{4 \cdot 6}{3} = 6 \end{aligned}$$

$$\Rightarrow \cos \theta = \frac{\mathbf{v} \cdot \mathbf{w}}{|\mathbf{v}| \cdot |\mathbf{w}|} = \frac{6}{6 \cdot 2} = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3} \text{ (or } 60^\circ)$$

(c) Find a unit vector $\mathbf{u}$ which has the same direction as $\mathbf{v}$.

$$\mathbf{u} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{1}{6}(3, \sqrt{3}, 2\sqrt{6}) = (\frac{1}{2}, \frac{\sqrt{3}}{6}, \frac{\sqrt{6}}{3})$$

(d) Find a number α so that $\mathbf{w} + \alpha\mathbf{v}$ is perpendicular to $\mathbf{v}$.

$$(\mathbf{w} + \alpha\mathbf{v}) \cdot \mathbf{v} = 0$$

$$\mathbf{w} \cdot \mathbf{v} + \alpha(\mathbf{v} \cdot \mathbf{v}) = 0$$

$$\alpha = -\frac{\mathbf{w} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} = -\frac{\mathbf{w} \cdot \mathbf{v}}{|\mathbf{v}|^2}$$

$$= -\frac{6}{6^2} = -\frac{1}{6}$$

Problem 2. (20 points)

- (a) Find an 2×2 matrix $\mathbf{A}$ such that $\mathbf{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + 2y \\ 2x - 3y \end{bmatrix}$.

$$\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix}$$

- (b) Find an 3×3 elimination matrix $\mathbf{E}$ that adds 4 times of row 1 to row 3.

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$$

- (c) Find an 3×3 permutation matrix $\mathbf{P}$ that exchanges rows 1 and 3, then rows 3 and 2.

$$\mathbf{P} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- (d) Find the inverse of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ if it is invertible.

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^{-1} &= \frac{1}{1 \cdot 4 - 2 \cdot 3} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} \\ &= -\frac{1}{2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \end{aligned}$$

Problem 3. (24 points)

(a) Find the LDU factorization of $A = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix}$

$$A = \begin{bmatrix} -1 & 1 & -3 \\ 2 & -4 & 1 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{\ell_{21}=-2} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ -3 & -3 & -4 \end{bmatrix} \xrightarrow{\ell_{31}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & -6 & 5 \end{bmatrix}$$

$$\xrightarrow{\ell_{32}=3} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix}$$

$$\Rightarrow A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & -3 \\ 0 & -2 & -5 \\ 0 & 0 & 20 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} -1 & & \\ & -2 & \\ & & 20 \end{bmatrix}}_D \underbrace{\begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & \frac{5}{2} \\ 0 & 0 & 1 \end{bmatrix}}_U$$

(b) Given $A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 2 \\ 2 & 2 & 3 \end{bmatrix}$, find A^{-1} using the Gaussian-Jordan Elimination.

$$[A: I] = \left[\begin{array}{ccc|ccc} \textcircled{1} & 1 & 2 & 1 & 0 & 0 \\ \textcircled{0} & 2 & 2 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} \textcircled{1} & 1 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ \textcircled{2} & 2 & 3 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 1 & \textcircled{2} & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & \textcircled{-1} & -2 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & \textcircled{0} & 0 & -3 & 0 & 2 \\ 0 & \textcircled{1} & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -2 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -1 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -2 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & -1 & 2 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 2 & 0 & -1 \end{array} \right]$$

$$\Rightarrow A^{-1} = \begin{bmatrix} 2 & -1 & 2 \\ -1 & 1 & 0 \\ 2 & 0 & -1 \end{bmatrix}$$

Problem 4. (20 points) Let $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 3 & 6 & 7 & 10 \end{bmatrix}$.

(a) Reduce A to the reduced row echelon form R .

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 8 & 10 \\ 3 & 6 & 7 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & -2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R.$$

(b) Identify the pivot and free variables, and then find the rank of A .

Pivot Variables: x_1, x_3
 Free variables: x_2, x_4 $r(A) = 2$

(c) Find the column space of A , $C(A)$.

$$C(A) = \left\{ c_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 8 \\ 7 \end{bmatrix} \mid c_1, c_2 \text{ any numbers} \right\}$$

(d) Find the null space of A , $N(A)$.

We need solve $RX = 0$, i.e.,

$$\begin{cases} x_1 + 2x_2 + x_4 = 0 \\ x_3 + x_4 = 0 \end{cases}$$

Let $x_2 = 1, x_4 = 0$, we get $S_1 = (-2, 1, 0, 0)$

$x_2 = 0, x_4 = 1$, we get $S_2 = (-1, 0, -1, 1)$

$$\Rightarrow N(A) = \left\{ c_1 S_1 + c_2 S_2 = c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} \mid c_1, c_2 \text{ any numbers} \right\}$$

$$= \begin{bmatrix} -2c_1 - c_2 \\ c_1 \\ -c_2 \\ c_2 \end{bmatrix}$$

Problem 5. (20 points) True or False (no explanation needed).

(a) If a matrix A is invertible, then A^2 must be invertible.

True $(A^2)^{-1} = (A^{-1})^2$

(b) The set of all vectors (b_1, b_2, b_3) with $b_1 \geq b_2 \geq b_3$ is a subspace of $\mathbb{R}^3$.

False $(2, 1, 1) \rightarrow (-2, -1, -1)$

(c) If a subspace of the vector space M contains $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$, then it must contain the identity matrix I .

True $\frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

(d) If A and B are symmetric, then their product AB must be symmetric.

False $(AB)^T = B^T A^T = BA \neq AB$ in general.
 $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$

(e) There exist two 2×2 matrices, A and B , such that $A \neq 0$ and $B \neq 0$ but $AB = 0$.

True. $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$